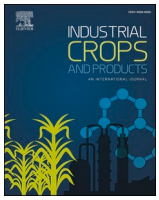
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Comparative metabolomics, network analysis, and fraction-based in vitro evaluation of three *Lagerstroemia* species reveal bioactive compounds and potential for sustainable industrial applications

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ABSTRACT

Lagerstroemia plants are widely cultivated as ornamental trees, while also being traditionally used in China and Southeast Asia as functional tea beverages, fragrant flowering trees, and herbal medicines, indicating their potential as industrial crops enriched with high-value bioactive compounds. However, insufficient understanding of how interspecific differences in metabolite composition has limited their broader application. Here, we performed metabolomics analysis on *Lagerstroemia speciosa* (LS), *Lagerstroemia caudata* (LC), and *Lagerstroemia indica* 'Dallas Red' (LI) by UPLC-MS/MS, detecting 2016 metabolites. Network-based analysis further identified 268 bioactive compounds, predominantly phenolics and triterpenoids, of which 39 potential core components were found to share common targets closely linked to type 2 diabetes–related therapeutic pathways. To validate their functional potential, crude extracts were fractionated and evaluated for phytochemical contents, antioxidant capacity, and α-glucosidase and α-amylase inhibitory activities, with *Eriobotrya japonica* (EJ) leaves used as an industrial reference. Although EJ showed higher total triterpenoids, several *Lagerstroemia* fractions exhibited stronger antioxidant capacity and α-glucosidase inhibition. Notably, the aqueous fraction of LS combined relatively high yield, abundant phenolics and flavonoids, strong antioxidant activity, and potent α-glucosidase inhibition. The aqueous fraction of LI showed similar functional potential, whereas LC displayed broader polarity-dependent bioactivity. Correlation analysis indicated that total phenolics, rather than total triterpenoids, were more closely associated with antioxidant and enzyme inhibitory activities. These results suggest that *Lagerstroemia* leaves are natural resources rich in water-soluble bioactive constituents, with potential for functional tea development and value-added utilization of ornamental industrial plant.

1. Introduction

Against the backdrop of pressure on high-quality biobased product resources, expanding the new applications of plants beyond their traditional roles can not only alleviate resource utilization constraints but also promote the diversified industrial development of functional plants (Caesar and Cech, 2019; M. Zhou et al., 2023). Genus *Lagerstroemia* L. stands as versatile plants, valued for their ornamental beauty, medicinal properties, environmental conservation, and cultural significance, all of which contribute to their significant industrial and economic potential (Yue et al., 2024). *Lagerstroemia speciosa* (LS), commonly known as Banaba, is traditionally renowned for its ability to regulate blood sugar (Bai et al., 2008). And its extracts are widely used in health supplements, additives, beverages, and pharmaceuticals, with major exports to the United States (US), Japan and Italy. Particularly, Japan imports thousands of tons of Banaba leaves annually from Philippines and Brazil to produce antidiabetic herbal teas, as one of its active ingredients, Corosolic acid, is highly regarded as natural insulin. In China and South Korea, Banaba extract has been included in the list of cosmetic ingredients. In addition, Banaba Leaf Extract Market size was

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0926-6690/© 2026 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC license (http://creativecommons.org/licenses/by-nc/4.0/).

valued at USD 338.60 Million in 2024 and the total revenue is expected to grow at a compound annual growth rate of 5.2% from 2025 to 2032, reaching nearly USD 507.94 Million (https://www.maximizemarketresearch.com/market-report/global-banaba-leaf-extract-market/77773/). *Lagerstroemia caudata* (LC) is rich in volatile monoterpenes, making it an important floral breeding resource within the genus. Its main floral scent components are geraniol (26.63%) and linalool (2.07%), which are highly valued for their antimicrobial, antioxidant, sedative, and antidepressant properties, laying the foundation for the potential industrial application of *L. caudata* as an essential oil–producing plant (Xu et al., 2024). In Chinese traditional medicine, the native plants *Lagerstroemia indica* (LI) (particularly their flowers, leaves, bark, and roots) were often made into decoctions or poultices for health purposes. Their applications in anti-aging, antiviral activity, and the treatment of eczema, boils, and carbuncles have been well documented (Yue et al., 2024). It is important to highlight that multiple *Lagerstroemia* plants exhibit remarkable antioxidant activities, with Banaba tea in particular demonstrating antioxidant capacity that has been shown to surpass that of green tea, oolong tea, black tea, and moringa tea (Afifah and Niwat, 2020).

According to data published by the International Diabetes Federation (IDF) (http://www.diabetesatlas.org/), the current global population of individuals with diabetes is approaching 600 million, with Type 2 Diabetes Mellitus (T2DM) accounting for more than 85%. Excessive production of reactive oxygen species in the body leads to oxidative stress, which is considered a key pathological mechanism in the onset, progression, and complications of diabetes (Hong et al., 2013; Li et al., 2013). Supplementing with external antioxidants is an effective strategy to prevent oxidative damage and reduce diabetes risk. In addition to food sources such as fruits, vegetables, grains, and oils, bioactive compounds extracted from natural plants are increasingly popular as dietary supplements among health-conscious consumers (Kim et al., 2019; Li et al., 2026). Although many synthetic drugs are effective in the prevention and treatment of diabetes, they are often associated with side effects such as weight gain and organ dysfunction. In contrast, extracts from *Lagerstroemia* species have consistently demonstrated good tolerability in various studies, with no reported adverse effects (Chang et al., 2023; Stohs et al., 2012).

Moreover, *Lagerstroemia* species have long growth periods and strong sprouting ability, characterized by rapid growth and fast biomass accumulation (Y. Zhou et al., 2023). During the growing season, their biomass increases substantially, and frequent pruning in summer is often required to maintain ornamental value, generating large amounts of horticultural waste and resulting in significant resource loss (Chang et al., 2023; Kim et al., 2019). These discarded plant materials are rich in bioactive compounds, such as high levels of phenolics and triterpenoids, which can serve as raw materials for industrial extraction and be used in the preparation of cosmetics, dietary supplements, and functional pharmaceutical ingredients. However, although the functional properties of eight *Lagerstroemia* species have been documented (Yue et al., 2024), their contemporary industrial and commercial applications remain underdeveloped. Therefore, *Eriobotrya japonica* (EJ) was introduced in this study as a reference tree species with a relatively mature basis for industrial application in China. EJ leaves are well known for their abundance of pentacyclic triterpenoid acids, such as ursolic acid, corosolic acid, and oleanolic acid, as well as flavonoids including kaempferol, quercetin and hyperoside. These functional constituents contribute to various biological activities of EJ leaves, including antioxidant, anti-inflammatory, anti-tumor, blood sugar lowering and α-glucosidase inhibitory effects, thereby supporting their industrial significance in traditional medicine and functional plant ingredient development (Dhiman et al., 2022; Zhu et al., 2022). Given that both EJ and *Lagerstroemia* leaves have been reported to possess triterpenoid acid enrichment and blood sugar lowering bioactivity potential, the inclusion of EJ as a reference species provides a practical benchmark for evaluating whether *Lagerstroemia* leaves could serve as complementary resources for the development of antidiabetic functional teas and health-related products.

However, comprehensive data on the large-scale identification, quantification, and comparison of metabolites across different *Lagerstroemia* species are lacking. Previous studies have only identified a limited number of compounds (Bai et al., 2008; Sandeep et al., 2019), and this narrow focus has constrained a more complete evaluation of the genus as a source of bioactive constituents. To obtain a thorough and profound understanding of the active components in *Lagerstroemia* plants and their potential mechanisms for diabetes treatment and hypoglycemic effects, this research applied the Ultra Performance Liquid Chromatography-Tandem Mass Spectrometry (UPLC-MS/MS) to systematically identify and relatively quantify metabolites in three typical *Lagerstroemia* species, including LS, LC and LI. Furthermore, a comprehensive analysis using network pharmacology conducted to identify the potential effective constituents from *Lagerstroemia* plants related to T2DM. In addition, in vitro assays preliminarily evaluated the antioxidant capacity and glucose-metabolism-related enzyme inhibitory activities of different solvent fractions. The findings support the sustainable use of *Lagerstroemia* by-products and their development into functional raw materials with antioxidant and antidiabetic properties, demonstrating the broader industrial relevance of ornamental plants for health promotion.

2. Materials and methods

2.1. Materials and reagents

We focused on the mature green leaves collected from three typical *Lagerstroemia* species (LS, LC, and LI) and *Eriobotrya japonica* (EJ) (Figure S1), with leaves harvested from different angles within the canopies of three selected plants for each species, grown under favorable conditions, and then mixed to form one sample. This process was replicated three times in total. Information regarding plant classification, collection sites, and plant photographs was presented in Table S1 and Figure S1.

The chemical reagents and drugs (analytical grade) used in this experiment included Folin-Ciocalteu reagent, sodium carbonate, sodium nitrite, sodium hydroxide, aluminum nitrate, vanillin, glacial acetic acid, perchloric acid, 2,2-diphenyl-1-picrylhydrazyl (DPPH), 2,2′-Azinobis(3-ethylbenzothiazoline-6-sulfonic acid) (ABTS), potassium persulfate (Macklin Biochemical, Shanghai, China), trolox, gallic acid, rutin, oleanolic acid, acarbose, α-glucosidase (Sigma-Aldrich, Shanghai, China), p-Nitrophenyl-α-D-glucopyranoside, α-amylase (Yuanye Bio-Tech, Shanghai, China), phosphate-buffered saline, anhydrous ethanol, chromatographic purity methanol, acetonitrile, ultrapure water, formic acid, petroleum ether, ethyl acetate, *n*-butanol, etc.

2.2. Widely-targeted metabolomic analysis on metabolites

Sample extraction, specific experimental parameters and operational methods followed the procedure previously established (Cao et al., 2023; Liu et al., 2024). Sample powder (50 mg) was extracted with 1200 μL of 70% methanol aqueous internal standard solution, vortexed repeatedly, centrifuged at 12,000 rpm for 3 min, filtered through a 0.22 μm membrane, and transferred to vials for subsequent analysis. A widely targeted metabolomics analysis was performed using a UPLC–ESI–MS/MS system equipped with an ExionLC™ AD UPLC system and a SCIEX tandem mass spectrometer. The detailed analytical conditions are listed in Table 1. All sample extracts were combined to prepare quality control (QC) samples, ensuring the reliability of the measurements. Qualitative analysis of substances was conducted based on the primary and secondary mass spectrometry data of metabolites, utilizing both the self-established MWDB (Metware Biotechnology Co., Ltd., Wuhan, China) and public databases (Chen et al., 2013; Li et al., 2022; Xia et al., 2023). Redundant features, including isotopic signals, K^+, Na^+,

Table 1
UPLC–MS/MS acquisition parameters for metabolite profiling.

Parameter	Details
LC Conditions	
Column	Agilent SB-C18 (1.8 μm, 2.1 mm × 100 mm)
Mobile Phase	A: Water (0.1% formic acid); B: Acetonitrile (0.1% formic acid)
Gradient Elution	0–9.00 min: linear increase of B from 5% to 95%; 9.00–10.00 min: hold at 95% B; 10.00–11.10 min: decrease B to 5%; 11.10–14.00 min: hold at 5% B for re-equilibration
Flow Rate	0.35 mL/min
Column Temperature	40 ℃
Injection Volume	2 μL
MS Conditions	
Ion Source	Electrospray ionization (ESI)
Source Temperature	500 ℃
Ion Spray Voltage (IS)	5500 V (positive mode); −4500 V (negative mode)
Gas Parameters	Curtain gas (CUR): 25 psi; Ion source gas 1 (GS1): 50 psi; Ion source gas 2 (GS2): 60 psi
Collision Gas	Nitrogen, medium
Collision-induced Dissociation	High
Scan Mode	Multiple reaction monitoring (MRM)
Data Acquisition	Declustering potential (DP) and collision energy (CE) were optimized for each MRM transition. Transitions were monitored in scheduled MRM mode according to analyte retention times.

and NH_4^+, as well as fragment ions originating from larger molecules, were excluded during data processing. Missing values were imputed using the K-nearest neighbors (KNN) algorithm, after which the coefficient of variation (CV) was calculated for each feature across QC samples. Features with CV values lower than 0.5 were retained to generate the final dataset. Metabolite quantification was achieved through the Multiple Reaction Monitoring (MRM) mode of triple quadrupole mass spectrometry.

2.3. Network pharmacology analysis

2.3.1. Prediction of potential active components

Network pharmacology analysis, as an innovative approach to drug discovery, was employed to decipher the intricate relationships between bioactive components and diseases (Cao et al., 2023; Li et al., 2022). The analytical flow of network pharmacology analysis was illustrated in Figure S2. Key active components were screened from 2016 metabolites using TCMSP (Traditional Chinese Medicine Systems Pharmacology Database and Analysis Platform, https://old.tcmsp-e.com/tcmsp.php), SwissADME (http://swissadme.ch/index.php), and previously reported literature (Castellano et al., 2013). Firstly, gastrointestinal absorption was set to "HIGH" as a criterion for drug absorption, screening for active compounds with high oral bioavailability. Secondly, compounds with "YES" in three or more of the species prediction rules (Lipinski, Ghose, Veber, Egan, Muegge) were considered active components, while those with "YES" in all five rules were identified as key active components of *Lagerstroemia* plants. Although certain components such as 1,2,3,4, 6-Penta-O-galloyl-β-D-glucose, Caffeic acid, Gallic acid, Casuarinin, Arjunic acid, Corosolic acid, Maslinic acid, 23-hydroxyursolic acid, Oleanolic acid, Ursolic acid, and Azelaic acid did not meet these two screening criteria, they had been reported in relevant studies to exhibit bioactivity. Therefore, they were also categorized as key active components to guarantee the identification of the most complete potential active components possible. Based on the selected key active components, SwissTargetPrediction (http://swisstargetprediction.ch/) was applied to predict possible targets, with targets having a probability > 0.12 selected for further analysis.

2.3.2. Prediction of core targets of Lagerstroemia plants and T2DM

Disease-related genes for "Type 2 Diabetes Mellitus" were retrieved from databases such as DisGeNET (https://www.disgenet.org/), OMIM (https://www.omim.org/), and GeneCards (https://www.genecards.org/) (Figure S2) (Cao et al., 2023). Once all targets were collected, duplicates were eliminated, and the names of the targets were standardized. A Venn diagram analysis was applied to identify shared targets between T2DM and the key active components of *Lagerstroemia* plants. These shared targets were uploaded into the STRING 11.5 database (https://cn.string-db.org/). Then, with Cytoscape 3.9.1, core targets were identified by selecting those with degree, closeness, and betweenness centrality scores above their respective averages. Moreover, the top 22 core targets of *Lagerstroemia* plants relevant to manage T2DM were predicted. Finally, based on the published literature and shared targets, the potential core active components for the treatment of T2DM were screened.

2.3.3. Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) analysis

To clarify the metabolic pathways and functions, the core target genes were uploaded into the David database (https://david.ncifcrf.gov/) for GO functional analysis and KEGG analysis. Then the GO and KEGG analysis results were visualized via the bioinformatics website (https://www.bioinformatics.com.cn/).

2.4. Total content analysis of phenolics, flavonoids and triterpenoids

Based on the previously reported method with some modifications (Kim et al., 2019). The dried powders of the four plant materials were ultrasonically extracted twice with 80% ethanol at a solid-to-liquid ratio of 1:20 for 30 min each time. The combined extracts were concentrated under reduced pressure by rotary evaporation to obtain the crude extracts. Subsequently, each crude extract was suspended in water and successively partitioned with petroleum ether, ethyl acetate, and *n*-butanol. The corresponding petroleum ether fractions (PE), ethyl acetate fractions (EA), *n*-butanol fractions (BuOH), and Water fractions (W) were obtained after solvent removal by vacuum rotary evaporation.

Total phenolics content was measured by the Folin–Ciocalteu method (Li et al., 2013) and expressed as gallic acid equivalents (GAE) per gram of dry weight (DW) powder of the sample (mg GAE/g DW). Total flavonoids were determined using the $NaNO_2$–$Al(NO_3)_3$ method (Dirar et al., 2019) and expressed as mg RE (rutin equivalents) /g DW. And total triterpenoids were quantified by the vanillin–perchloric acid method (Wang et al., 2021) and expressed as mg OE (oleanolic acid equivalents) /g DW.

2.5. In vitro antioxidant capacity of three Lagerstroemia species and E. japonica

The DPPH and ABTS radical scavenging activity of the sample was measured through published research methods with some modifications (Dirar et al., 2019). The radical scavenging capacity was expressed as mg TE (Trolox equivalents)/g DW.

2.6. α-glucosidase and α-amylase inhibition assay

The α-glucosidase and α-amylase inhibitory activity was determined using 96 microplate method with gradient-diluted extracts, with acarbose as the positive control (Chang et al., 2023; Tian et al., 2021). The inhibition ratio (%) was calculated according to the formula: Inhibition ratio (%) = $[1 - (OD_{test} - OD_{blank}) / (Control\ OD_{test} - Control\ OD_{blank})] \times 100$

2.7. Data analysis

All experimental data were statistically analyzed using SPSS 22 (IBM

SPSS, Armonk, NY, USA) and are expressed as mean ± standard deviation (SD) (n = 3). Two-way analysis of variance (ANOVA) was performed to evaluate the effects of species, fraction type, and their interaction. Different lowercase letters indicate significant differences among all samples based on one-way ANOVA followed by Tukey's multiple comparison test ($p < 0.05$). Multivariate statistical analysis techniques were employed to process metabolomics data. Principal Component Analysis (PCA) was prompted by statistical functions in R (https://www.r-project.org). Prior to unsupervised PCA, unit variance scaling was applied to the data. Hierarchical Cluster Analysis (HCA) of metabolites was performed using the R package ComplexHeatmap and presented as a heatmap with a dendrogram. The supervised Orthogonal Partial Least Squares Discriminant Analysis (OPLS-DA) model in the R package Metabo Analyst R was used to identify differential metabolites. Graphics were created using the Metware Cloud platform (https://cloud.metware.cn), Chiplot (https://www.chiplot.online/), Origin software (Version 2022, OriginLab Corporation, Northampton, MA, USA) and Adobe Illustrator (Version 2022, Adobe Inc., San Jose, CA, USA).

3. Results

3.1. Metabolic profiling of three Lagerstroemia plants

To comprehensively determine the metabolic profiles of the three species of *Lagerstroemia*, UPLC-MS/MS was employed to analyze three *Lagerstroemia* plants (LS, LC, LI). The overlap display of total ion current (TIC) and the multi-peak detection diagram of QC samples were shown in Figure S3. The observed overlap rate of TIC curves in the QC samples was quite high (Figure S3 A-B), providing strong evidence for assessing the reproducibility and reliability of the results. In the multi-peak MRM metabolite detection diagram (Figure S3 C-D), each identified metabolite was indicated by a distinctively colored mass spectrometry peak. Furthermore, empirical cumulative distribution function (ECDF) plot based on coefficient of variation (CV) and the Pearson correlation analysis between QC samples further support the good technical repeatability of the test results (Figure S4 A-B).

A total of 2016 primary and secondary metabolites were identified from the three *Lagerstroemia* plants (Table S2, Fig. 1 A), with 1780 metabolites in LS, 1837 in LC, and 1889 in LI (Table S2, Figure S5). These metabolites could be classified into flavonoids (497, 24.65%), phenolic acids (268, 13.29%), amino acids and their derivatives (208, 10.32%), lipids (201, 9.97%), terpenes (155, 7.69%), organic acids (122, 5.95%), alkaloids (97, 4.81%), lignans and coumarins (85, 4.22%), nucleotides and their derivatives (70, 3.47%), tannins (73, 3.62%), quinones (22, 1.09%), steroids (3, 0.15%), and others (217, 10.76%). A correlation analysis of metabolites from different *Lagerstroemia* plant samples (Fig. 1B) revealed that LC and LI exhibited relatively similar secondary metabolite patterns ($0.43 \leq |r| \leq 0.50$), likely due to their closer genetic backgrounds (Dong et al., 2021; Y. Zhou et al., 2023).

3.2. Multivariate statistical analysis and overall metabolic difference

Unsupervised principal component analysis (PCA) was conducted on the samples to obtain an initial understanding of the overall metabolite variations among three *Lagerstroemia* species, as well as the extent of differences among biological replicates (Fig. 1 C). The tight clustering of QC samples on the PCA score plot validated the consistency and dependability of the chromatography and mass spectrometry systems. Notably, the PCA score plot demonstrated a pronounced separation trend among the groups, indicating significant differences between the three species of *Lagerstroemia*. Furthermore, the tight grouping of the three biological replicates within each species demonstrated excellent reproducibility among groups. The cumulative variance explained by the two principal components (PC1 and PC2) amounted to 79.21%, with PC1 and PC2 accounting for 46.04% and 33.17%, respectively. Although LC and LI showed less separation along PC1, the three sample groups

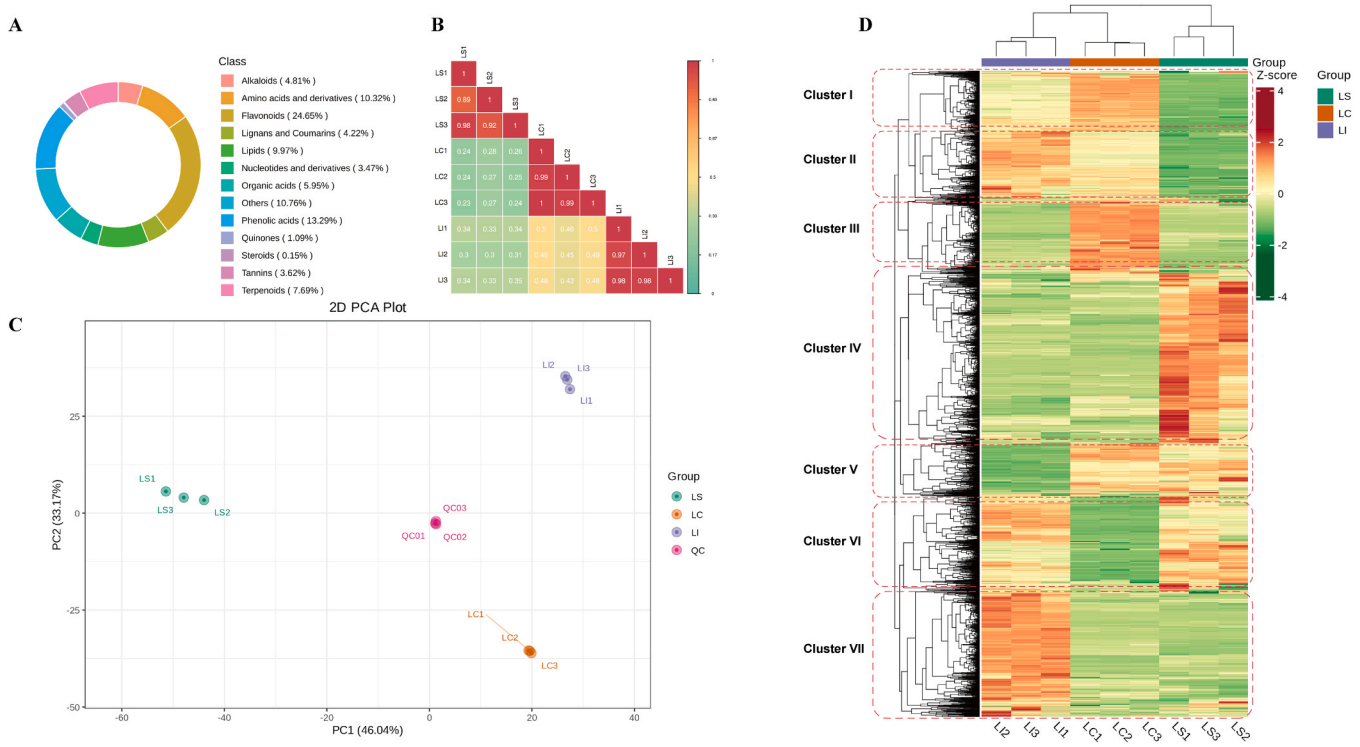


Fig. 1. (A) Categorization of the 2016 metabolites of three *Lagerstroemia* plants. (B) Correlation analysis. (C) PCA score plot, with the following color codes for the sampling groups: green, LS; red, LC; purple, LI; pink, QC. (D) HCA of metabolites. The normalized signal intensities of metabolites were depicted through a color spectrum, with red denoting high levels and green indicating low level.

displayed distinct profiles of metabolites overall.

Moreover, hierarchical cluster analysis (HCA) was carried out using the relative abundance of metabolites (Fig. 1D). The analysis distinctly indicated that metabolites could be successfully clustered into seven subgroups, with Clusters II and VII showing the highest levels in LI, Clusters I and III in LC, and Cluster IV peaking in LS. In line with the PCA results, this study revealed that the metabolite profiles of each group were significantly distinct.

Orthogonal partial least squares discriminant analysis (OPLS-DA) was utilized to refine the identification of distinct metabolites across pairwise comparisons of LS, LC, and LI. (Fig. 2). For these pairwise comparisons, both the Q^2 and R^2Y values surpassed the threshold of 0.9 (Figure S6), suggesting robust stability and dependability of the models. According to the OPLS-DA model score plots (Figure S6), a significant separation could be observed among different *Lagerstroemia* species, demonstrating the existence of distinct metabolome profiles. Metabolites exhibiting differences were identified using the criteria of VIP ≥ 1 and Fold change ≥ 2 or ≤ 0.5, and visualized through volcano plots (Fig. 2).

During pairwise analyses comparing LC versus (vs.) LS, LI vs. LC, and LI vs. LS, there were respectively 1065, 889, and 1045 significantly different metabolites identified (Table S3, Fig. 2). It indicated pronounced differences in the makeup and regulation of metabolites among distinct species. Additionally, K-means clustering was employed to group the relative abundances of distinctive metabolites sourced from screenings across all subgroups. All notably different metabolites were categorized into three major classes and nine subclasses (Figure S7). Specifically, 525, 410, and 537 metabolites exhibited higher levels in LS, LC, and LI, respectively.

3.3. Network pharmacology analysis

3.3.1. Prediction of active compounds and core targets

By integrating the TCMSP database, SwissADME, and published literature, numerous metabolites with promising health benefits in *Lagerstroemia* plants were successfully identified. Comprehensive analysis yielded a total of 268 key active compounds, including 109 flavonoids, 29 lignans and coumarins, 24 terpenes, 22 alkaloids, 19 phenolic acids, 9 tannins, and others (Table S4). Among these, 246 potential key active compounds and their corresponding 16,370 protein targets (after removing the value of 0) were screened in the SwissTargetPrediction database, while the remaining 22 components had no corresponding targets. Deduplication of the 16,370 protein targets resulted in 1170 unique targets (Table S5). Additionally, 1524 T2DM targets were retrieved from OMIM (taking the first 519 targets), DisGeNET (taking the first 1000 targets), and GeneCard (taking the first 1000 targets) databases after removing duplicates (Table S5). Venn diagram analysis revealed 258 shared targets between *Lagerstroemia* plants and T2DM (Fig. 3 A, Table S5). Of the 268 key active compounds, 26 show no shared targets with T2DM (Table S4).

To gain deeper insights into the importance of *Lagerstroemia* plants in targeting T2DM, the 258 shared T2DM targets were evaluated using STRING 11.5 database and Cytoscape 3.9.1 software. Ultimately, 22 core targets for *Lagerstroemia* plants and T2DM were identified, with a Degree value greater than 100 serving as the selection criterion (Table S5). These targets played pivotal roles in T2DM. For instance, Quercetin significantly reduced inflammation by inhibiting the production of TNF-α and IL-6 and suppressing NF-κB pathway activity, thereby improving insulin resistance (Salehi et al., 2020). In endothelial cells treated with IL-6 and LPS, Chalcones were found to eliminate inflammation by inhibiting the activation of STAT3 and NF-κB. Furthermore, triterpenes like Oleanolic acid and Ursolic acid enhanced insulin signaling and lipid metabolism through AKT1 and PPARG activation while alleviating inflammation by inhibiting NFKB1(Castellano et al., 2013). Consequently, *Lagerstroemia* plants primarily treated diabetes by modulating the expression levels of proteins involved in various pathways, including TNF, AKT1, IL-6, and PPARG, providing valuable evidence for its potential therapeutic applications.

3.3.2. Go and KEGG enrichment analysis

To better elucidate the mechanism of *Lagerstroemia* plants in treating T2DM, GO and KEGG pathway enrichment analyses were performed on 258 intersecting targets using DAVID. The analyses revealed 1008, 111, and 200 entries associated with BP, CC, and MF, respectively (Table S6), with the top 10 entries listed in Fig. 3B. Notably, among the prominently enriched BP, we found the positive modulation of MAPK cascades, protein phosphorylation, and gene expression regulation, hinting that *Lagerstroemia* plants potentially harness their therapeutic potential by fine-tuning cellular signaling cascades, modulating protein modifications, and influencing gene expression patterns. For CC, significant mentions encompass the plasma membrane, receptor complexes, and membrane rafts, suggesting a primary focus of *Lagerstroemia* plants on these membrane-bound structures and their associated components. This targeting likely facilitates the regulation of signal transduction mechanisms and substance transportation processes. Regarding MF, the enrichment encompassed protein kinase activities, enzyme bindings, and ATP bindings, which implies that *Lagerstroemia* plants may regulate intracellular energy metabolism and signal transduction dynamics by exerting control over the functionality of key enzymes within the cellular machinery.

For instance, extracts from medicinal plants like *Scutellaria baicalensis* Georgi could activate the MAPK signaling pathway, enhancing cellular antioxidant enzyme expression, boosting antioxidant capacity, and mitigating oxidative stress-induced cellular damage (Hong et al., 2013). Similarly, catechins in green tea bind to cellular membranes, affecting membrane fluidity and receptor complex function, thereby modulating cellular signal transduction (Singh et al., 2011). Additionally, *Panax ginseng* C.A.Mey. extracts have been proven to activate AMPK kinase, enhancing intracellular energy metabolism and promoting glucose uptake and utilization (Kang et al., 2017). Analogously, extracts of *Lagerstroemia* plants may regulate BP across multiple cellular compartments and functions to treat T2DM, which helps explain its traditional use as an antidiabetic tea.

Furthermore, the analysis identified 187 signaling pathways related to T2DM (Table S6), with the 20 most prominent pathways depicted in Fig. 3 C. They were predominantly associated with pathways linked to cancer, diabetes and its complications, atherosclerosis, and cardiovascular diseases. The AGE-RAGE signaling pathway played a pivotal role in T2DM by mediating the binding of advanced glycation end products to their receptor RAGE, activating downstream signaling, and triggering oxidative stress and inflammation, processes intimately linked to diabetic complications such as neuropathy and nephropathy (Chen et al., 2020). Effective active compounds in *Lagerstroemia* plants could impact pathways associated with diabetes and cardiovascular diseases, underscoring the promising potential in diabetes prevention and treatment.

3.3.3. Prediction of potential core bioactive compounds for T2DM

Earlier research has indicated the hypoglycemic effects of leaf extracts from LS and LI, yet relatively few related active components have been identified (Chang et al., 2023). This study thoroughly investigated the possible pharmacological components of three *Lagerstroemia* species for treatment of T2DM through network pharmacology, with 242 out of 268 key active components exhibiting shared targets withT2DM (Table S4). Subsequently, based on published literature and screening of core targets, 39 core active components with hypoglycemic activity were finally identified in the three *Lagerstroemia* species (Table S7), far exceeding the currently known hypoglycemic components of *Lagerstroemia*. It contained 20 flavonoids, 7 pentacyclic triterpenes, 3 phenolic acids, 2 tannins, 2 amino acids and their derivatives, and 5 other components (Table S7). Among them, flavonoids and pentacyclic triterpenes emerged as the primary pharmacological components,

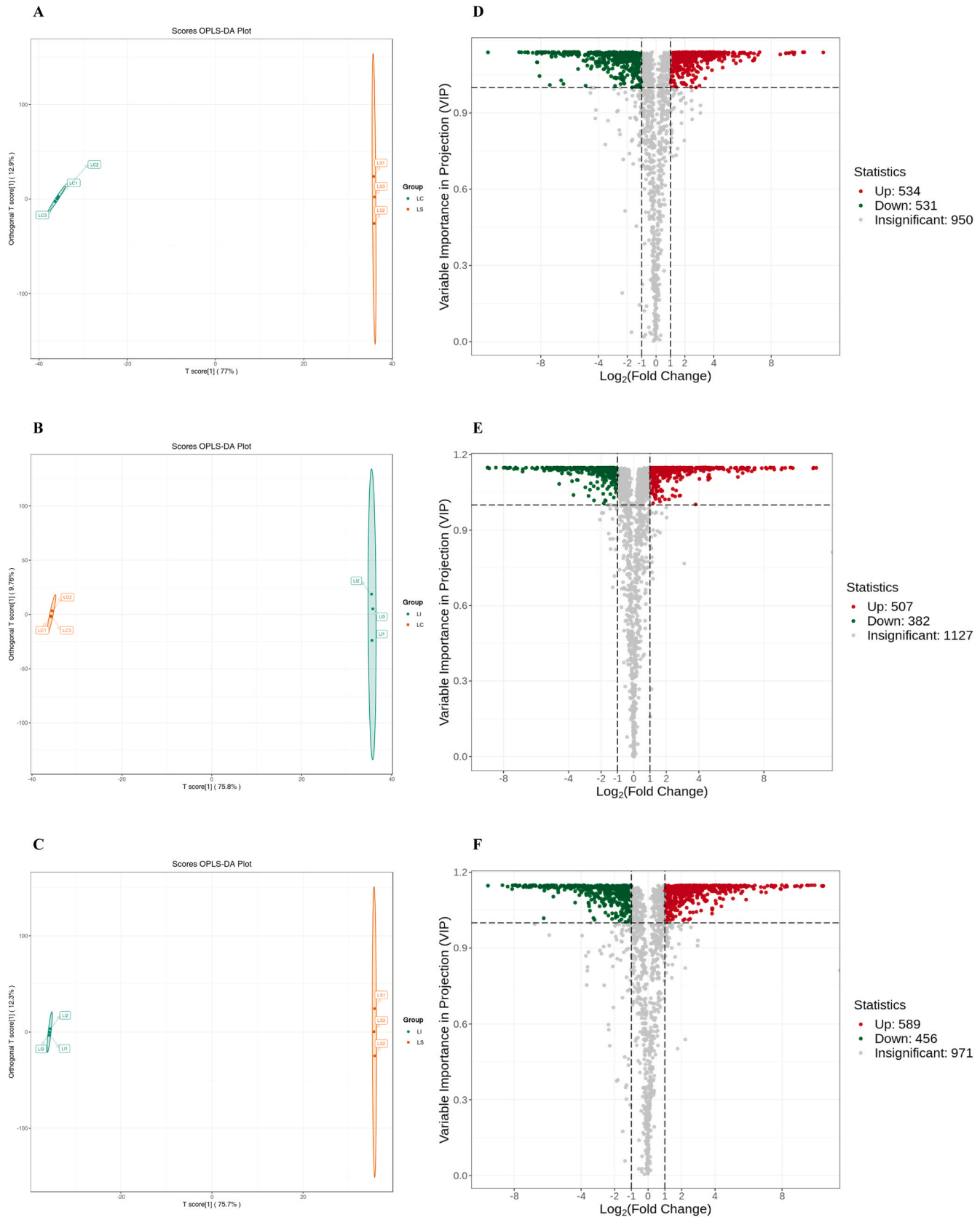


Fig. 2. (A-C) The pairwise comparison score plots generated by OPLS-DA for differential metabolites. (A) LC vs. LS; (B) LI vs. LC; (C) LI vs. LS. (D-F) Volcano plots that visualize the expression levels of differential metabolites. (D) LC vs. LS; (E) LI vs. LC; (F) LI vs. LS.

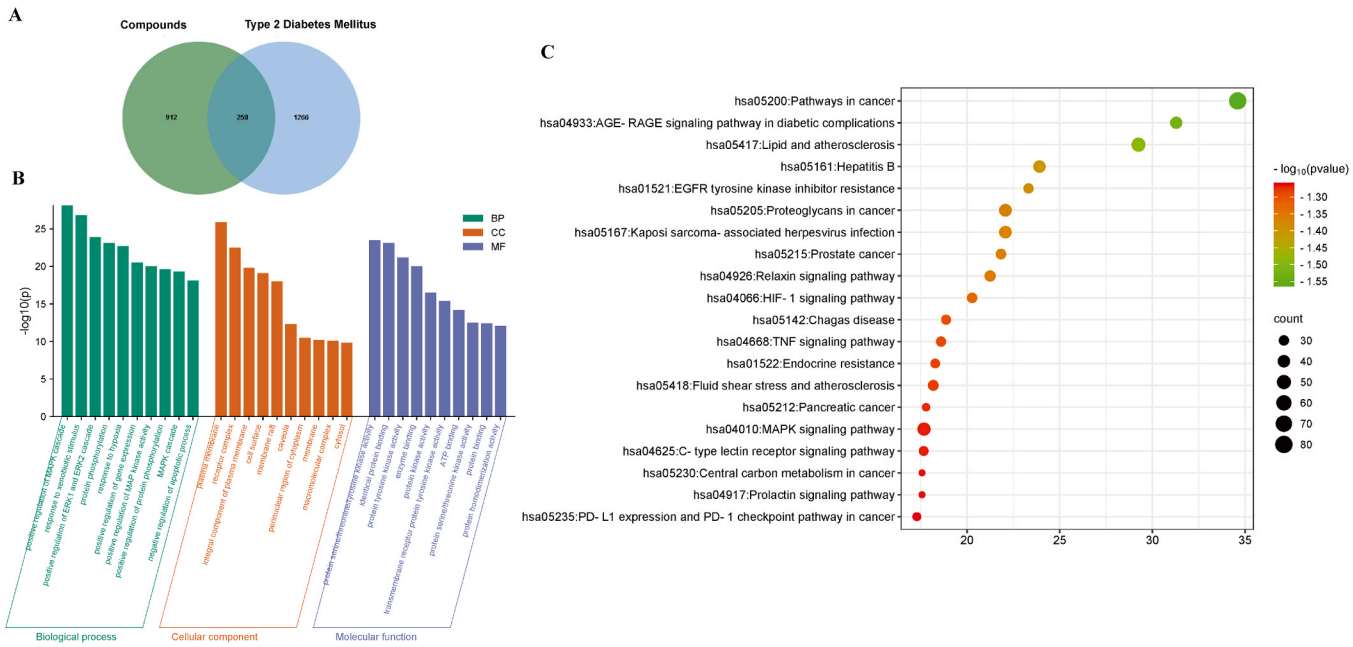


Fig. 3. (A) Venn diagram of shared targets between *Lagerstroemia* components and T2DM. (B) The top 10 entries of GO enrichment analysis. (C) The 20 most prominent pathways of KEGG enrichment analysis.

followed by phenolic acids, tannins, and others.

These 39 components have been shown in previous *in vitro*, *in vivo*, and clinical studies to possess multiple effects, including regulating blood glucose levels, improving insulin resistance, alleviating diabetic complications, stimulating glucose consumption, and reducing inflammation (Table S7). The relative contents of these components in three species were depicted in Figure S8. Previous research has identified Asiatic acid, Corosolic acid, 1,2,3,4,6-Penta-O-galloyl-β-D-glucose, 3-O-Methylellagic acid, and Ellagic acid as the hypoglycemic marker compounds in LS (Bai et al., 2008). Among these, all except Corosolic acid exhibited higher relative expression levels in LS. However, other potential diabetes-treating triterpenoids belonging to the pentacyclic triterpenoid class, such as betulinic acid, ursolic acid, maslinic acid, 23-hydroxyursolic acid, and oleanolic acid (Qian et al., 2021), were more abundant in LC and LI, potentially indicating the potential of other *Lagerstroemia* species as hypoglycemic plants.

3.4. Differential accumulation of key bioactive compounds in three Lagerstroemia plants

To elucidate the differences in the 268 bioactive compounds among the three *Lagerstroemia* species, we conducted a more systematic and detailed analysis. This analysis clarifies the chemical fingerprint characteristics of each species and provides a theoretical basis for targeted compound selection and differentiated functional utilization.

3.4.1. Phenols

Research has indicated a close relationship between oxidative stress and metabolic disorders like diabetes and obesity. The potent antioxidant and anti-inflammatory properties of phenols could mitigate oxidative stress and inflammation-induced damage to pancreatic islet cells, thereby improving insulin secretion and function. A total of 133 phenolic bioactive compounds were identified through network pharmacology analysis, including 105 flavonoids, 19 phenolic acids, and 9 tannins. Fig. 4A illustrates the accumulation patterns of these compounds across the three *Lagerstroemia* species. Among them, LI exhibited the lowest relative abundance of phenolic compounds, while LS and LC showed comparable levels, which is likely due to the lower accumulation of flavonoids in LI. The PLS-DA score plot (Fig. 4B) further demonstrated clear distinctions in phenolic profiles among the three species. The top 15 differential metabolites and their corresponding VIP scores derived from this analysis are shown in Fig. 4C. Moreover, hierarchical clustering analysis (Fig. 4D) revealed significant differences in the composition and abundance of phenolic bioactive compounds across the three *Lagerstroemia* species.

As the predominant phenolic constituents, flavonoids can exert antidiabetic effects while preserving cellular homeostasis, offering a safer alternative to synthetic antidiabetic agents (Tanveer et al., 2017). For instance, the relative concentrations of various flavonoids such as Naringenin, Genistein, Genkwanin, Syringetin, Pinobanksin, Acacetin, Luteolin, Irigenin, Galangin, Apigenin, Isovitexin, Chrysin, and Cajanin were higher in LI than in LC and LS. These components have been confirmed to possess multiple potential pharmacological activities. Naringenin, a high-concentration flavanone compound found in citrus fruits, has also been detected in *Lagerstroemia* leaves, which was confirmed to exhibit anti-inflammatory, antioxidant, immunomodulatory, nephroprotective, hepatoprotective, and antidiabetic properties. By upregulating AMPK and inhibiting gluconeogenesis, Naringenin exerts a hypoglycemic effect similar to metformin (the only oral hypoglycemic drug to date associated with a reduction in macrovascular complications in diabetic patients), but without the known side effects of metformin like diarrhea due to local lactic acid accumulation (Nyane et al., 2017). Furthermore, Apigenin was a versatile compound effective in treating various diseases across multiple organ systems, including diabetes, and promoting oral and skin health (Allemailem et al., 2024). Potential bioactive flavonoids with higher relative content in LC were Phloretin, Didymin, Kaempferol, Kumatakenin, Nevadensin, Poncirin, Carthamone, and others. For example, Phloretin, a dihydrochalcone, was a natural phenolic compound that was implicated in activating the expression of apoptosis-related genes and signaling within molecular pathways, with antidiabetic, antioxidant, anti-cancer, anti-allergic and hepatoprotective effects (Mariadoss et al., 2019). On the other hand, flavonoids with higher relative content in LS were observed for Nobiletin, Quercetin, 3-Methylkaempferol, Epicatechin, Maesopsin, Pinostrobin, Cyrtominetin, Nepetin, and Azaleatin. Among them, Quercetin and its derivatives have been extensively studied for their anti-diabetic, antioxidant, anti-inflammatory, anti-cancer, cardiovascular, and wound healing effects, with dietary intake or supplementation sufficient to

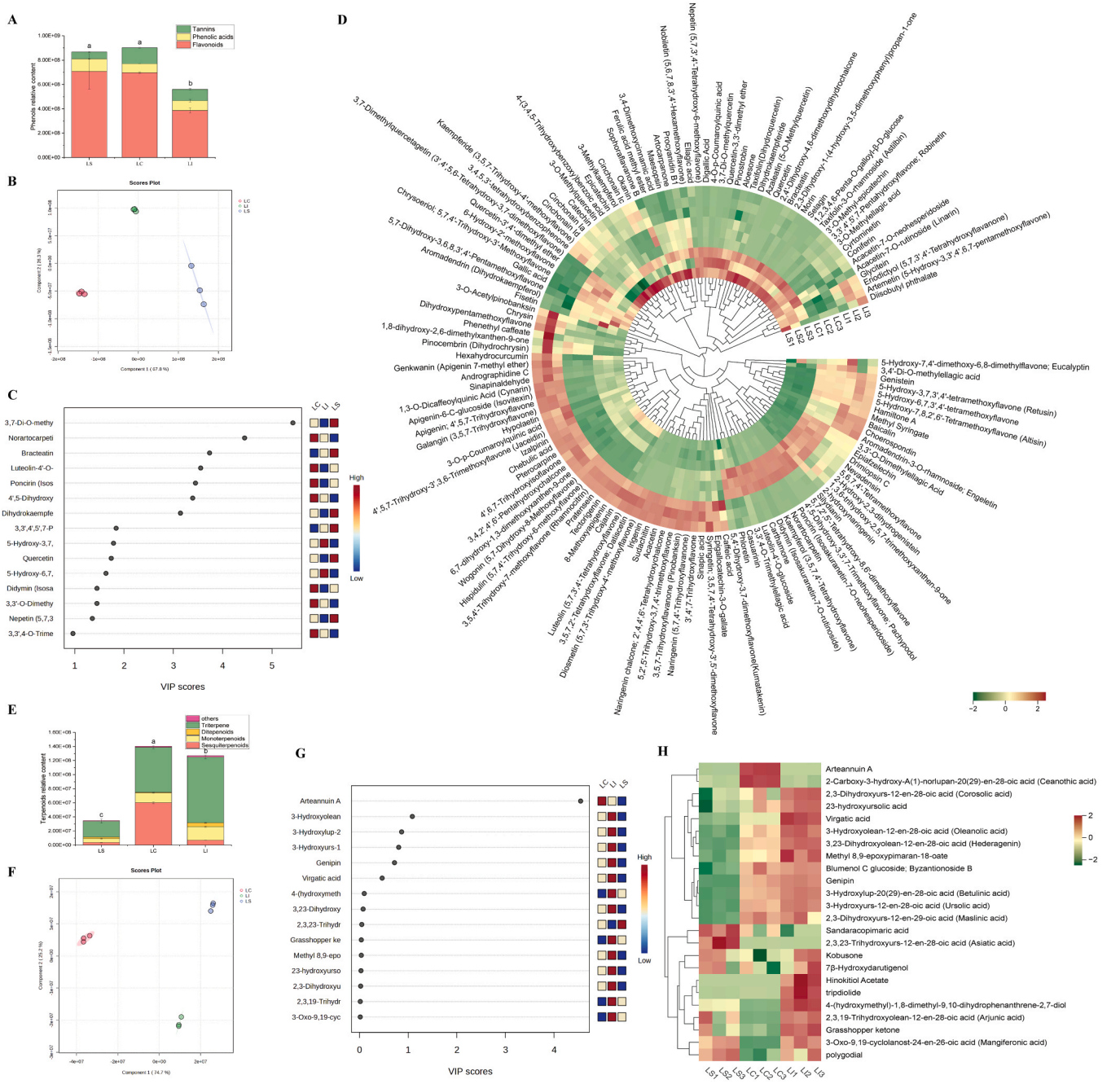


Fig. 4. Figures A–D show the accumulation patterns, PLS-DA score plots, VIP score plots and hierarchical clustering of bioactive phenolic compounds among the three *Lagerstroemia* species, while Figures E–H present the corresponding analyses for bioactive terpenoid compounds.

confer protective benefits (Salehi et al., 2020). These compounds likely underlie the traditional medicinal uses of *Lagerstroemia* species in addressing a broad spectrum of health conditions.

In addition, we observed that relatively abundant tea polyphenols were detected in the three *Lagerstroemia* plants, including Catechin, Epicatechin, Epicatechin Gallate, Epigallocatechin, Epigallocatechin Gallate, Epigallocatechin-3-O-gallate and Gallocatechin. These compounds are widely present in traditional teas and have strong biological activities such as antioxidant, anti-inflammatory and hypoglycemic effects (Khan and Mukhtar, 2018). Their presence in *Lagerstroemia* plants not only enriches the functional component spectrum of these plants, but also provides a material basis for their development as natural and healthy industrial products.

Phenolic acids, another crucial class of natural phenolic compounds in *Lagerstroemia*, are renowned for their strong antioxidant activity. For instance, Hexahydrocurcumin, Cynarin, Coniferin, Sinapinaldehyde, 3-O-p-Coumaroylquinic acid, Caffeic acid, and Sinapic acid were abundant in LI, while Aloesone, 4-O-p-Coumaroylquinic acid, Gallic acid, 1,2,3,4,6-Penta-O-galloyl-β-D-glucose, Ferulic acid methyl ester, and 3,4-Dimethoxycinnamic acid were enriched in LS. Several of these compounds have been studied for their profound effects on neurodegenerative diseases like Parkinson's and Alzheimer's (Elsawi et al., 2018). In addition, tannins such as 3-O-methylellagic acid, Casuarinin, and Ellagic acid, which have been associated with the hypoglycemic and anti-obesity effects of LS (Bai et al., 2008; Yue et al., 2024), were also detected in LI and LC.

3.4.2. Terpenoids

Terpenoids are essential for defense mechanisms of plants against stress, insects, and microorganisms, while their diverse biological activities hold immense potential for human applications, encompassing medicinal uses, fragrances, as well as food and industrial applications (Xu et al., 2024). Among the three detected *Lagerstroemia* species, a total of 155 terpenoid compounds were identified (Table S2), which aligns with previous research, emphasizing the abundance of terpenes in *Lagerstroemia* (Yue et al., 2024).

In Fig. 4E-H, we present the 24 potential bioactive terpenoids identified through network pharmacology analysis, including 3 sesquiterpenes, 3 diterpenes, 4 tetraterpenes, and 12 triterpenes. Fig. 4E illustrates the accumulation patterns of these compounds across the three *Lagerstroemia* species. Overall, LC exhibited the highest relative abundance of bioactive terpenoids, whereas LS showed the lowest, which is likely attributable to the substantial enrichment of sesquiterpenes in LC. Notably, LI displayed higher enrichment of bioactive triterpenoids, suggesting that other *Lagerstroemia* species may possess functional properties similar to those of LS. The PLS-DA score plot (Fig. 4F) further revealed clear distinctions in terpenoid profiles among the three species. The top 15 differential metabolites and their VIP scores selected from this analysis are shown in Fig. 4G. Moreover, hierarchical clustering analysis (Fig. 4H) confirmed the significant differences in the composition and abundance of bioactive terpenoids among the three *Lagerstroemia* species.

Previous studies have demonstrated that triterpenoids could comprehensively prevent diabetes and exhibit diverse anti-tumor, anti-inflammatory, antioxidant, cytoprotective, and antiviral activities (Qian et al., 2021). And Corosolic acid has been identified as the hallmark hypoglycemic compound in LS (Bai et al., 2008). However, surprisingly, the relative content of Corosolic acid was actually lower in LS among the three *Lagerstroemia* species, which suggested that LI and LC may possess more significant levels of Corosolic acid, indicating great potential for commercial value exploration. Interestingly, when comparing the relative contents of structural analogs of Corosolic acid, such as Asiatic acid, Oleanolic acid, Ursolic acid, Maslinic acid, and Betulinic acid, it was found that Asiatic acid was more abundant in LS, while the others were present in higher concentrations in LI and LC. This finding challenged the traditional view that Corosolic acid was the biomarker for hypoglycemic effects in LS, suggesting that Asiatic acid may be another key component contributing to its blood glucose-lowering properties (Kim et al., 2019). Furthermore, research has shown that the bioavailability of Asiatic acid was markedly higher than that of Corosolic acid, Ursolic acid, and Oleanolic acid (Qian et al., 2021), making it a potentially promising candidate for further research and development.

3.4.3. Other bioactive components

In addition to the major phenolic and terpenoid constituents, other classes of bioactive compounds were also identified, including amino acids and their derivatives, organic acids, alkaloids, lipids, lignans, and

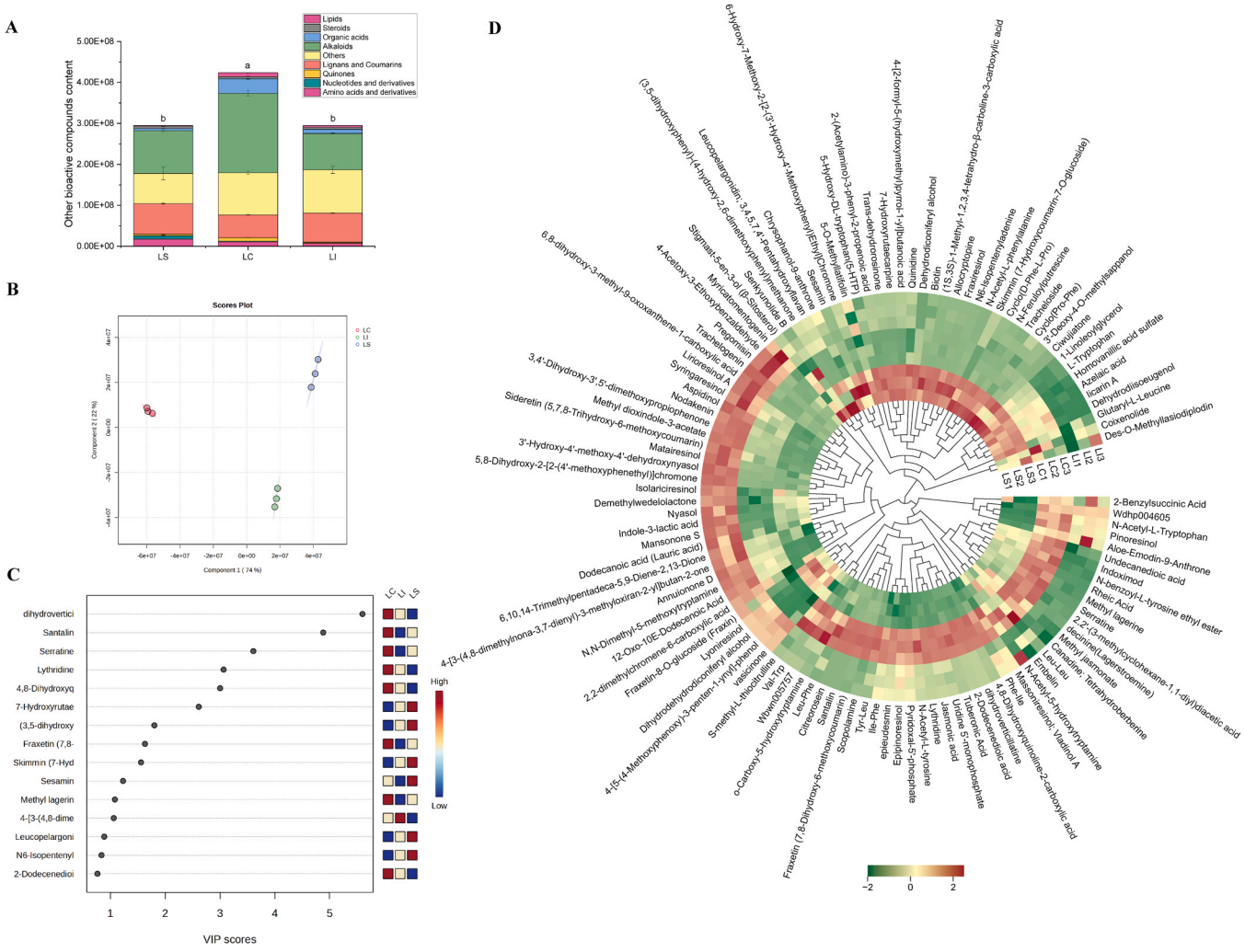


Fig. 5. Figures A–D show the accumulation patterns, PLS-DA score plots, VIP score plots and hierarchical clustering of other bioactive compounds (excluding phenolics and terpenoids) among the three *Lagerstroemia* species.

coumarins. The accumulation patterns of these compounds across the three *Lagerstroemia* species are shown in Fig. 5A. Among the three *Lagerstroemia* species, LC had the highest relative content of active compounds, while LI and LS had comparable contents, largely attributed to the content of active alkaloids in LC. The PLS-DA score plot (Fig. 5B) demonstrates that the composition of bioactive compounds can effectively discriminate among the three species; Fig. 5C presents the top 15 differential metabolites selected from this analysis along with their VIP scores. Hierarchical clustering analysis (Fig. 5D) further indicates significant differences among the three species. In addition to their health benefits, these compounds may also influence the flavor and taste of *Lagerstroemia* herbal tea, including amino acids contributing to umami, sweetness, and bitterness, organic acids responsible for acidity, and sugars that enhance aroma and sweetness. To some extent, this supports the traditional use of LS as a health-promoting functional beverage.

3.5. Fraction yield and contents of total phenolics, flavonoids, and triterpenoids in three Lagerstroemia species and E. japonica

As can be seen from Fig. 6, there are obvious differences in the contents of various indicators in different extraction phases of the four plant materials. Two-way ANOVA revealed highly significant effects of species, fraction type, and their interaction on fraction yield and the contents of total phenolics, flavonoids, and triterpenoids ($P < 0.001$). As shown in Fig. 6A, the aqueous fraction generally exhibited high yields across all four plant materials, suggesting that these leaves contained abundant water-soluble or highly polar constituents. However, the yield distribution patterns among fractions differed markedly among species. Specifically, LS was mainly enriched in the aqueous fraction, LI showed relatively high yields in the aqueous and *n*-butanol fractions, LC showed higher yields in the aqueous and petroleum ether fractions, and EJ exhibited higher yields in the aqueous and ethyl acetate fractions.

Among all samples, the petroleum ether fractions showed the lowest total phenolic content, indicating limited enrichment of phenolic compounds in the low-polarity fraction (Fig. 6B). In contrast, the aqueous and *n*-butanol fractions of the three *Lagerstroemia* species generally contained relatively high levels of total phenolics. The aqueous fraction of LS exhibited the highest total phenolics, reaching 43.13 ± 1.02 mg GAE/g DW, which was significantly higher than that of all other samples, followed by the aqueous fraction of LI. In LC, total phenolics were more evenly distributed, with relatively high levels detected in the ethyl acetate, *n*-butanol, and aqueous fractions. Unlike the *Lagerstroemia* species, phenolic compounds in EJ were mainly enriched in the ethyl acetate and *n*-butanol fractions, whereas the aqueous fraction showed lower total phenolics. These results suggest that the polarity-dependent distribution pattern of phenolic compounds in EJ leaves differed markedly from that observed in the *Lagerstroemia* species.

Similar to total phenolics, the petroleum ether fractions showed the lowest total flavonoid content among all plant materials (Fig. 6C). Among the three *Lagerstroemia* species, the aqueous fraction of LS had

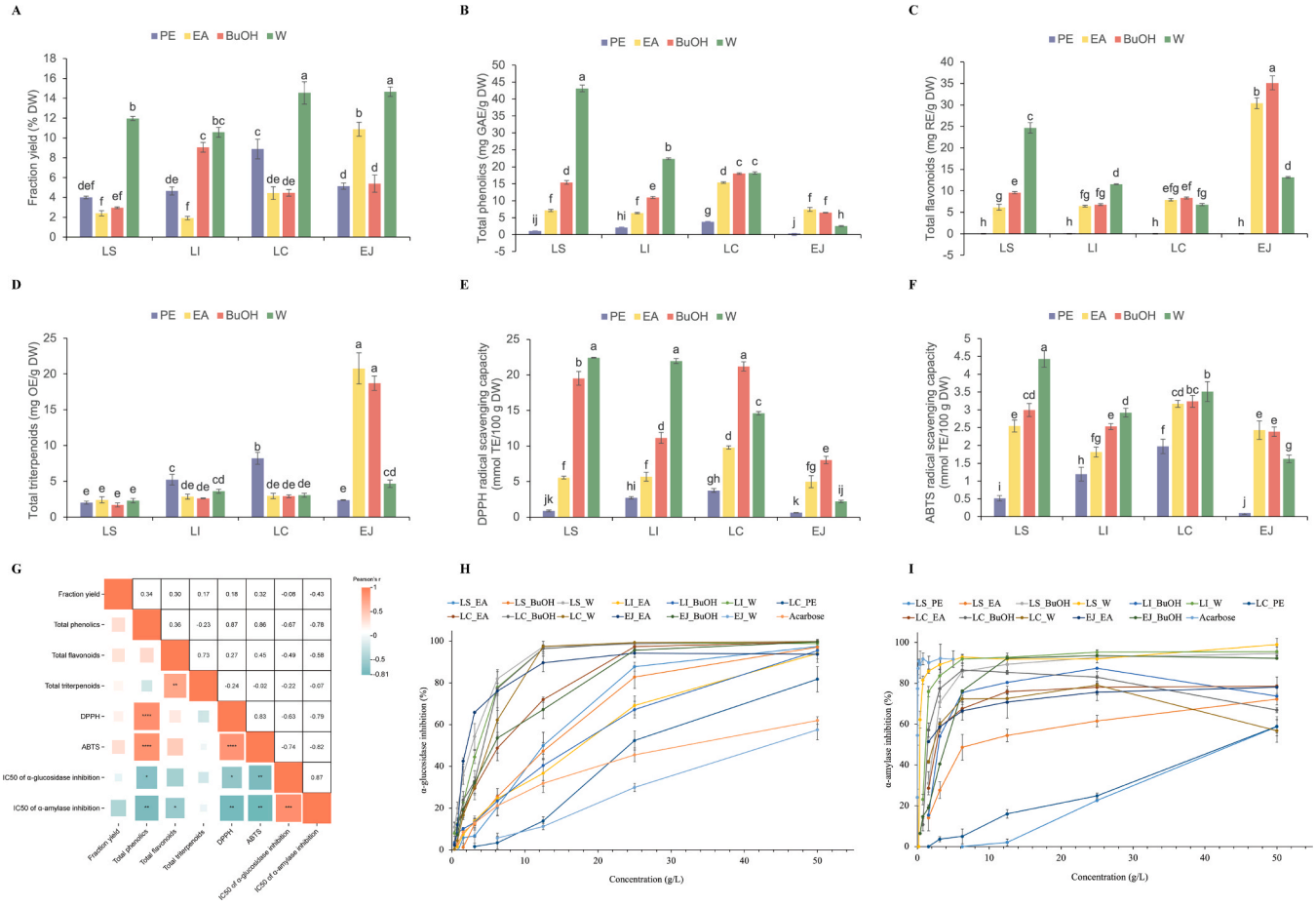


Fig. 6. Fraction yield (A), total phenolic content (TPC, B), total flavonoid content (TFC, C), total triterpenoid content (TTC, D), DPPH radical scavenging capacity (E), ABTS radical scavenging capacity (F), Pearson correlation analysis among the measured parameters (G), dose-response curves of α-glucosidase inhibition (H), and dose-response curves of α-amylase inhibition (I) of different fractions from *Lagerstroemia speciosa* (LS), *Lagerstroemia caudata* (LC), *Lagerstroemia indica* 'Dallas Red' (LI), and *Eriobotrya japonica* (EJ). PE, petroleum ether fraction; EA, ethyl acetate fraction; BuOH, *n*-butanol fraction; W, aqueous fraction. Data are presented as mean ± SD. Different lowercase letters indicate significant differences among samples ($p < 0.05$). For IC_{50}-related analyses, samples without measurable IC_{50} values within the tested concentration range were excluded from statistical comparison and correlation analysis.

the highest total flavonoids (24.67 ± 1.22 mg RE/g DW), significantly exceeding those of the other *Lagerstroemia* fractions. However, EJ exhibited significantly higher total flavonoid levels than the *Lagerstroemia* species, particularly in the ethyl acetate (30.38 ± 1.27 mg RE/g DW) and *n*-butanol fractions (35.12 ± 1.62 mg RE/g DW), suggesting that EJ leaves are rich in flavonoids and polyphenols of medium to relatively high polarity.

Among the three *Lagerstroemia* species, LS showed relatively low total triterpenoid contents across all fractions, ranging from 1.69 to 2.40 mg OE/g DW, with no significant differences among fractions. In contrast, LI and LC exhibited a more pronounced enrichment in the petroleum ether fraction. The petroleum ether fraction of LC had the highest total triterpenoid content among the *Lagerstroemia* samples (8.19 ± 0.83 mg OE/g DW), suggesting that LC may contain higher levels of low-polarity or lipophilic triterpenoid constituents (Fig. 6D). Compared with the *Lagerstroemia* species, EJ showed markedly higher total triterpenoid contents, particularly in the ethyl acetate (20.77 ± 2.17 mg OE/g DW) and *n*-butanol fractions (18.70 ± 1.00 mg OE/g DW). This result supports the rationality of using EJ as a reference species for relatively mature corosolic acid and triterpene acid resources in China.

3.6. Antioxidant capacity of three Lagerstroemia species and E. japonica

Significant differences in DPPH radical scavenging capacity were observed among the fractions of the four plant materials (Fig. 6E). Overall, the *Lagerstroemia* species showed markedly stronger DPPH antioxidant capacity than EJ, particularly in the aqueous and *n*-butanol fractions of LS, LI, and LC. The antioxidant activity of the *Lagerstroemia* species was mainly distributed in the aqueous and *n*-butanol fractions, whereas that of EJ was primarily enriched in the ethyl acetate and *n*-butanol fractions. The aqueous fraction of LS exhibited the highest DPPH scavenging capacity (22.41 ± 0.05 mmol TE/100 g DW), which was in the highest significance group together with the aqueous fraction of LI and the *n*-butanol fraction of LC. In contrast, EJ showed generally lower DPPH scavenging capacity across all fractions, with the *n*-butanol fraction being the highest among them, reaching only 8.03 ± 0.51 mmol TE/100 g DW.

The distribution pattern of ABTS radical scavenging capacity was similar to that of DPPH (Fig. 6F). The aqueous fraction of LS showed the highest ABTS scavenging capacity (4.43 ± 0.23 mmol TE/100 g DW), significantly exceeding all other samples. The aqueous, *n*-butanol, and ethyl acetate fractions of LC also exhibited relatively strong ABTS scavenging capacities, indicating that the antioxidant activity of LC was distributed across medium to high polarity fractions. Overall, the petroleum ether fractions showed the lowest activities in both DPPH and ABTS assays, suggesting that low-polarity constituents may contribute less to the radical scavenging capacity of these plant materials.

3.7. α-glucosidase and α-amylase inhibitory activity of three Lagerstroemia species and E. japonica

The α-glucosidase and α-amylase inhibition rates at the highest tested concentration and the corresponding IC_{50} values are summarized in Table 2, while the dose–response curves are presented in Fig. 6H-I to illustrate the concentration-dependent inhibitory patterns of the fractions. All concentrations were expressed as plant dry weight equivalents rather than extract dry weight, to facilitate subsequent analyses with phytochemical contents and antioxidant capacities measured on the same plant dry weight basis.

For α-glucosidase inhibition, most fractions exhibited high inhibitory rates at the highest tested concentration of 50 g/L, except for some petroleum ether fractions and the aqueous fraction of EJ. The IC_{50} values varied markedly among plant species and fractions. The aqueous fraction of LS showed the strongest α-glucosidase inhibitory activity, with an IC_{50} value of 2.194 ± 0.253 g/L, which was significantly lower than that of most other fractions. The ethyl acetate fraction of EJ also exhibited strong activity, with an IC_{50} value of 2.462 ± 0.278 g DW equivalent/L, comparable to that of the LS aqueous fraction. The aqueous fraction of LI, the *n*-butanol fraction of LC, and the aqueous fraction of LC also showed relatively strong α-glucosidase inhibition. In contrast, the petroleum ether fractions generally showed weaker activity. The petroleum ether fractions of LS, LI, and EJ did not reach 50% inhibition within the tested concentration range, and therefore their IC_{50} values could not be determined. Notably, the positive control acarbose had an α-glucosidase IC_{50} value of 29.074 ± 3.531 g/L, which was markedly higher than those of several plant fractions under the present assay conditions. The extracts of leaves from the three *Lagerstroemia* species exhibited significantly stronger inhibitory activity than acarbose, aligning with previous studies on the flower extracts of *L. indica* and *L. indica* f. *alba* (Chang et al., 2023), which suggests that different species and tissues of *Lagerstroemia* may possess strong in vitro α-glucosidase inhibitory potential.

The α-amylase inhibition results were not completely consistent with those of α-glucosidase. Overall, several fractions also showed strong α-amylase inhibitory activity, but clear differences were observed among plant species and fractions. At the highest tested concentration of

Table 2
Inhibitory effects of different fractions from four plant species against α-glucosidase and α-amylase.

	α-glucosidase inhibition at 50 g/L (%)	IC_{50} of α-glucosidase inhibition (g/L)	α-amylase inhibition at 50 g/L (%)	IC_{50} of α-amylase inhibition (g/L)
LS_PE	8.15 ± 5.56	> 50	58.72 ± 3.01	42.712 ± 1.489 h
LS_EA	97.43 ± 0.87	11.113 ± 1.389 e	72.25 ± 2.85	11.255 ± 2.673 g
LS_BuOH	97.09 ± 3.02	11.238 ± 1.175 e	94.58 ± 0.97	1.982 ± 0.510 de
LS_W	99.74 ± 1.05	2.194 ± 0.253 a	98.96 ± 3.01	0.434 ± 0.028 b
LI_PE	25.05 ± 5.00	> 50	35.01 ± 1.79	> 50
LI_EA	94.03 ± 4.29	13.578 ± 2.801 e	29.40 ± 2.48	> 50
LI_BuOH	95.38 ± 0.78	12.741 ± 1.287 e	87.39 ± 1.01	3.659 ± 0.835 f
LI_W	98.81 ± 1.42	2.904 ± 0.338 abc	95.56 ± 2.26	0.860 ± 0.221 c
LC_PE	81.73 ± 6.07	25.336 ± 3.151 f	58.94 ± 4.70	43.441 ± 3.804 h
LC_EA	99.51 ± 0.23	5.127 ± 0.674 d	78.61 ± 0.59	2.957 ± 0.404 ef
LC_BuOH	99.78 ± 0.70	3.339 ± 0.277 bc	83.02 ± 2.69	1.788 ± 0.222 de
LC_W	99.66 ± 0.74	4.097 ± 0.199 cd	74.89 ± 6.20	1.971 ± 0.090 e
EJ_PE	2.69 ± 2.36	> 50	18.95 ± 0.44	> 50
EJ_EA	93.69 ± 2.21	2.462 ± 0.278 ab	78.10 ± 4.95	1.187 ± 0.134 cd
EJ_BuOH	99.61 ± 0.24	5.284 ± 1.090 d	92.29 ± 2.56	3.441 ± 0.526 f
EJ_W	57.49 ± 4.77	42.644 ± 4.565 g	38.82 ± 6.56	> 50
Acarbose	61.93 ± 1.78	29.074 ± 3.531 fg	93.61 ± 0.86	0.013 ± 0.003 a

Note: Inhibition rates at the highest tested concentration (50 g/L) and IC_{50} values of α-glucosidase and α-amylase inhibition are shown for different fractions from *Lagerstroemia speciosa* (LS), *Lagerstroemia caudata* (LC), *Lagerstroemia indica* 'Dallas Red' (LI), and *Eriobotrya japonica* (EJ). PE, petroleum ether fraction; EA, ethyl acetate fraction; BuOH, *n*-butanol fraction; W, aqueous fraction. Values are presented as mean ± SD. Different lowercase letters within the same column indicate significant differences among samples ($p < 0.05$). >50, not detectable because the inhibition rate did not reach 50% at the highest tested concentration.

50 g/L, most fractions showed relatively high α-amylase inhibition except for the petroleum ether and ethyl acetate fractions of LI, and the petroleum ether and aqueous fractions of EJ, which did not reach 50% inhibition within the tested range. The aqueous fraction of LS showed the highest inhibition rate, reaching 98.96 ± 3.01%. Based on IC_{50} values, the aqueous fraction of LS exhibited the strongest α-amylase inhibitory activity (IC_{50} = 0.434 ± 0.028 g/L), significantly lower than those of the other plant fractions. The aqueous fraction of LI also showed strong activity, with an IC_{50} value of 0.860 ± 0.221 g DW equivalent/L. The ethyl acetate fraction of EJ, the *n*-butanol fraction of LS and LC, and the aqueous fraction of LC also exhibited relative high inhibitory activity, with IC_{50} values of 1.187 ± 0.134, 1.982 ± 0.509, 1.788 ± 0.222, and 1.971 ± 0.090 g/L, respectively. In contrast to the α-glucosidase results, acarbose showed a much lower α-amylase IC_{50} value (0.013 ± 0.003 g/L) than all plant fractions, indicating that acarbose had substantially stronger in vitro α-amylase inhibitory activity.

3.8. Pearson correlation coefficient analysis

Pearson correlation analysis was performed to evaluate the relationships among the measured parameters. As shown in the correlation heatmap (Fig. 6G), fraction yield showed relatively weak correlations with most bioactivity-related parameters, indicating that extraction yield alone does not directly reflect the abundance of active constituents or the strength of biological activity.

In contrast, total phenolics was strongly positively correlated with DPPH and ABTS radical scavenging capacities, with correlation coefficients of r = 0.87 and r = 0.86, respectively, both reaching a highly significant level. This suggested that phenolic compounds may be the major contributors to the antioxidant activity of the different fractions. A strong positive correlation was also observed between DPPH and ABTS values (r = 0.83), indicating strong consistency between the two antioxidant assays and both methods effectively reflected the variation in radical scavenging capacity among samples.

Total flavonoids content was strongly correlated with total triterpenoids (r = 0.73), suggesting that flavonoids and triterpenoids may be co-enriched in some fractions. However, the correlations between total flavonoid content and DPPH or ABTS radical scavenging capacities were relatively weak, with correlation coefficients of r = 0.27 and r = 0.45, respectively. This indicates that flavonoids may contribute to the antioxidant activity, but their overall contribution may be lower than that of total phenolics.

Total phenolics were negatively correlated with $\log_{10}(IC_{50})$ values for both α-glucosidase and α-amylase inhibition (r = −0.67 and −0.78, respectively), while DPPH and ABTS showed similar negative correlations, with ABTS showing the strongest association with α-amylase $\log_{10}(IC_{50})$ (r = −0.82). Since lower IC_{50} values indicate stronger inhibitory activity, these negative correlations suggest that fractions with higher total phenolics and stronger antioxidant capacity generally tended to exhibit stronger α-glucosidase and α-amylase inhibitory activities. In addition, $\log_{10}(IC_{50})$ values for α-glucosidase and α-amylase inhibition were strongly positively correlated (r = 0.87), suggesting that samples with stronger α-glucosidase inhibition also appeared to show stronger α-amylase inhibitory activity.

Although EJ exhibited markedly higher total triterpenoids, the relatively weak correlations between total triterpenoids and the inhibitory activities against α-glucosidase and α-amylase indicated that enzyme inhibition in these fractions was not primarily determined by the total triterpenoids level. Moreover, in this preliminary in vitro screening, the antioxidant and enzyme inhibitory activities of EJ were not consistently superior to those of the *Lagerstroemia* species. These findings suggested distinct functional resource characteristics between EJ and *Lagerstroemia*. EJ may serve as a representative triterpenoid-rich reference species, whereas *Lagerstroemia* plants, particularly LS, may be more appropriately positioned as water-extracted, phenolic-rich plant resources with potential applications in tea beverages and water-based functional products.

4. Discussion

As the boundary between ornamental and functional plants continues to blur, the multifunctional development and sustainable industrial utilization of ornamental plant resources are receiving increasing attention (Tang et al., 2025; Yue et al., 2024; M. Zhou et al., 2023). Exploring their bioactive potential has become an emerging trend in horticultural science. Natural products have a wide range of sources, including animals, plants, microorganisms, minerals, and even humus (Melnychuk et al., 2024; Sawarkar et al., 2023). Regarding plant sources, in addition to conventional natural functional sources such as fruits, vegetables, and grains, many horticultural plants and their byproducts are also rich in natural bioactive compounds, which possess diverse beneficial properties, such as antioxidant, antidiabetic, anti-obesity, and anti-aging effects (Caesar and Cech, 2019; Yue et al., 2024; M. Zhou et al., 2023). Traditionally, beautiful *Lagerstroemi*a plants have been valued primarily for landscape enhancement and garden ornamentation. However, *Lagerstroemia speciosa* (L.) Pers. (LS) has a long history as a traditional tea for controlling blood sugar and losing weight, and has gained popularity as a health drink in East Asia and the United States (Yue et al., 2024). Furthermore, the health supplement made from LS extract, Corosolic acid, has obtained the Japanese functional labeling food registration license and has passed three clinical pharmacology evaluations for the treatment and management of T2DM and obesity. These findings highlight the potential of *Lagerstroemia* species as valuable multifunctional resources and provide a start point for further exploration of their high-value development and diverse applications.

It has been shown that widely targeted metabolomics is increasingly employed to identify the metabolite profiles of medicinal, ornamental, and tea plants. For instance, Peng et al. (2024) identified 1076 compounds in *Campanumoea javanica*, X. Cao et al. (2023) detected 1126 compounds in citrus herbs, H. Liu et al. (2024) identified 1099 compounds in *Ziziphus jujube* leaf black tea, and Y. Hou et al. (2023) reported 1248 metabolites in *Camellia sinensis* var. assamica. Compared to traditional analytical methods, this approach significantly enhances identification efficiency. Through UPLC-MS/MS analysis, we found that three *Lagerstroemia* species (LS, LI and LC) showed significant differences in metabolite compositions, and a total of 2016 primary and secondary metabolites identified. Notably, in three *Lagerstroemia* species, metabolomics has revealed the presence of hallmark tea compounds such as Catechin, Epicatechin, Epigallocatechin, Epigallocatechin-3-O-gallate and so on. Additionally, a diverse array of amino acids, organic acids, sugars, polyphenols, and flavonoids, known for their impact on herbal tea aroma, flavor, and color, has been detected. More importantly, compounds with central nervous system stimulant activity, like caffeine, theophylline, and theobromine, were absent in the these plants (Hou et al., 2023; Khan and Mukhtar, 2018), indicating that *Lagerstroemia*-based tea may possess similar flavor and nutritional properties to traditional tea while offering a milder profile, making it a suitable option for health-conscious consumers seeking natural, gentle products.

Subsequent network pharmacology analysis of the 2016 compounds indicated that the active ingredients contributing to functional effects in *Lagerstroemia* plants should not be solely attributed to Corosolic acid, as flavonoids, phenolic acids, tannins, Corosolic acid structural analogs and other active substances may also play important roles. The 39 identified potential core active components (Table S7) have been found to exhibit potential in treating T2DM and its related complications by inhibiting α-glucosidase, regulating insulin levels, lowering blood sugar and triglycerides, reducing oxidative stress and improving kidney function (Bai et al., 2008; Qian et al., 2021; Yue et al., 2024). These results provided a theoretical basis for the subsequent selection of total phenolics, total flavonoids, total triterpenoids, antioxidant capacity, and glucose-lowering enzyme inhibitory activity as validation indicators.

The fractionation experiment further revealed the functional performance of these potential active constituents in fractions with different polarities. Overall, the aqueous fractions of the *Lagerstroemia* species generally showed relatively high yields, indicating that their leaves contain abundant water-soluble or highly polar constituents. In particular, the aqueous fraction of LS not only exhibited a relatively high extraction yield, but also showed the highest or relatively high levels of total phenolics, total flavonoids, DPPH and ABTS radical scavenging capacities, and the strongest α-glucosidase inhibitory activity. These results are consistent with the abundant phenolic acids and flavonoid metabolites detected by widely targeted metabolomic analysis and support the material basis of LS for the development of Banaba tea and water-extracted functional industrial products. Moreover, previous studies have also reported that when assessing TPC and antioxidant activity in various traditional herbal teas, LS tea exhibited the highest TPC and antioxidant activity, even surpassing green tea (*Camellia sinensis* (L.) Kuntze), with TPC reaching 20.54 ± 0.31 mg GAE/g and showing the highest DPPH (11.19 ± 0.12 μmol TE/g) and Ferric Reducing Antioxidant Power (FRAP) activity (0.66 ± 0.02 mmol $FeSO_4$/g) (Afifah and Niwat, 2020).

LC showed a broader polarity-dependent distribution pattern. Its aqueous, *n*-butanol, and petroleum ether fractions exhibited distinct advantages in fraction yield, antioxidant activity, enzyme inhibitory activity, and total triterpenoids, respectively, suggesting its potential for the integrated development of active constituents across multiple polarity ranges. The aqueous fraction of LI also showed relatively high total phenolics and strong enzyme inhibitory activity, indicating that this commonly cultivated ornamental species also has potential as a functional plant resource.

Notably, the widely targeted metabolomic results showed that LC and LI had higher relative levels of total triterpenoids and several representative triterpenoids (corosolic acid, ursolic acid, oleanolic acid, maslinic acid, and betulinic acid) than LS. This pattern was consistent with the total triterpenoids measured in the polarity-based fractions, further indicating that LC and LI may have similar application potential as a source of corosolic acid as LS. Murthy et al. (2020) found that TTC in LS could reach 166 mg/g using HPLC, with Corosolic acid being the most notable triterpenoid, ranging from 0.05 to 8.90 mg/g DW (Lin et al., 2023). HPTLC analysis confirmed that Corosolic acid content could reach up to 12.87 ug/mg in ethanol extracts of LS leaves (Thakur and Devaraj, 2020). Previous studies have mainly focused on the triterpenoids of LS, but few have directly compared it with other triterpenoids-rich plant resources. In the present study, EJ was included as a reference species. The ethyl acetate and *n*-butanol fractions of EJ showed significantly higher total triterpenoid contents than those of the three *Lagerstroemia* species. Compared with EJ, a well-established industrial reference resource for corosolic acid and related triterpenoids, the *Lagerstroemia* species may have less prominent industrial value as triterpenoids-rich resources.

Notably, the correlation analysis in this study showed that total triterpenoid content was only weakly correlated with DPPH and ABTS radical scavenging capacities, as well as with α-glucosidase and α-amylase inhibitory activities, whereas total phenolic content showed strong correlations with these bioactivity-related parameters. This suggested that *Lagerstroemia* plants have complementary functional value distinct from that of EJ. Moreover, these findings were consistent with the widely targeted metabolomic results, which revealed abundant phenolic acids and flavonoids, and with the network pharmacology analysis, in which phenolic acids and flavonoids were predicted as important candidate bioactive compounds. Overall, the functional activities of *Lagerstroemia* plants may not be determined by corosolic acid alone, but rather by the combined contributions of multiple classes of constituents, including phenolic acids, flavonoids and certain triterpenoids.

In addition, the α-glucosidase inhibition results showed that the medium to high polarity fractions of the four plant materials exhibited stronger inhibitory activity than acarbose. This is consistent with previous studies reporting that various plant-derived polyphenols and triterpenoids, such as corosolic acid, oleanolic acid, and maslinic acid, possess effective α-glucosidase inhibitory activity. In contrast, the plant fractions showed relatively milder α-amylase inhibition than acarbose. For functional teas or products intended to support postprandial blood glucose management, strong α-glucosidase inhibition combined with moderate α-amylase inhibition is generally considered desirable (Tan et al., 2017), because excessive α-amylase inhibition may be associated with incomplete carbohydrate digestion and an increased risk of gastrointestinal discomfort (Lam et al., 2024). However, it should be noted that crude plant fractions contain complex mixtures of polyphenols, flavonoids, tannins, and triterpenoids, which may produce strong apparent inhibitory effects through multi-site binding, synergistic interactions, or even nonspecific protein interactions (Caesar and Cech, 2019; Vidar et al., 2023). Therefore, these in vitro results should not be directly regarded as evidence that the fractions have stronger hypoglycemic effects than acarbose in vivo. Further studies involving enzyme kinetics, cell-based assays, validation of individual compounds, and in vivo experiments are needed to confirm their physiological relevance.

Overall, this study supports the industrial development potential of *Lagerstroemia* plants from multiple perspectives, including metabolite composition, network-based prediction, fractionation-based validation, and functional activity evaluation. Compared with EJ, which serves as an industrial reference species enriched in triterpenoids, *Lagerstroemia* plants may be more appropriately positioned as water-extracted functional tea resources and high-value materials derived from ornamental plant by-products. In particular, the aqueous fraction of *L. speciosa* combined a relatively high extraction yield with abundant water-soluble phenolics and flavonoids, strong antioxidant capacity, and significant α-glucosidase inhibitory activity, highlighting its potential for the development of functional teas, water extracts, and health-promoting plant ingredients. However, this study remains at the stage of in vitro screening. Further studies on compound-level validation, synergistic mechanisms, in vivo efficacy, safety, and product standardization are needed to support future application.

Abbreviations

ABTS, 2,2′-azinobis(3-ethylbenzothiazoline-6-sulfonic acid; BP, Biological Process; CC, Cellular Component; CV, coefficient of variation; DPPH, 2,2-diphenyl-1-picrylhydrazyl; DW, dry weight; EJ, *Eriobotrya japonica* (Thunb.) Lindl.; GAE, gallic acid equivalents; GO, Gene ontology; HCA, Hierarchical Cluster Analysis; KEGG, Kyoto encyclopedia of genes and genomes; LC, *Lagerstroemia caudata* Chun & F.C.How ex S.K.Lee & L.F.Lau; LI, *Lagerstroemia indica* 'Dallas Red'; LS, *Lagerstroemia speciosa* (L.) Pers.; MF, Molecular Function; MRM, Multiple Reaction Monitoring; OE, Oleanolic acid equivalents; OPLS-DA, Orthogonal Partial Least Squares Discriminant Analysis; PCA, Principal Component Analysis; QC, quality control; RE, rutin equivalents; TE, trolox equivalents; T2DM, Type 2 Diabetes Mellitus. TFC, total flavonoid content; TIC, total ion current; TPC, phenolic content; TTC, total triterpenoid content; UPLC-MS/MS, Ultra Performance Liquid Chromatography-Tandem Mass Spectrometry; US, United States.

CRediT authorship contribution statement

Ziwei Yue: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yanan Guo:** Writing – review & editing, Methodology, Investigation. **Jingyun Wang:** Writing – review & editing, Investigation, Conceptualization. **Yan Xu:** Writing – review & editing, Methodology. **Ming Cai:** Writing – review & editing, Visualization, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Donglin Zhang:** Writing – review & editing, Supervision. **Qixiang Zhang:** Writing – review & editing, Supervision.

Authorship

We certify that all authors have made significant contributions to this manuscript and confirm that the manuscript has been read and approved by all authors. The author confirms that the order of authors listed in the manuscript has been approved by all authors.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.indcrop.2026.124377.

Data Availability

Data will be made available on request.

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